ZCAS UNIVERSITY

BScNE21- NETWORK ENGINEERING

CIT2351 NETWORK DESIGN AND SECURITY

FINAL EXAMINATIONS

TUESDAY, 27th MAY, 2025

TIME: 14:00 – 17:00 HOURS

TIME ALLOWED: THREE (03) HOURS

INSTRUCTIONS TO CANDIDATES:

1. **SECTION A:** This section is **compulsory** with **ONE (01)** Question and carries **40 marks.**
2. **SECTION B:** Attempt **any THREE (03)** questions from this section. Each question carries **20 marks**.
3. This examination paper carries a total of **100 marks.**
4. Time allowed: **THREE (03) Hours plus FIVE (05) minutes to read through the paper.**
5. **DO NOT** turn this page **until** you are told to do so.

SECTION A – COMPULSORY QUESTIONS

INSTRUCTIONS: You MUST attempt **ALL QUESTIONS** that follow the case study provided.

QUESTION ONE

CASE STUDY: *Kopala University*

A new university, ***Kopala University*** has just been opened in Kalulushi, Copperbelt Province. The University has 150 members of staff including its faculty employees. They have approached you to design a network that ensures security, scalability, and efficient resource management and further to propose security measures to ensure it security.

a) Describe the main phases of network design per the TOP-DOWN approach. **(06 marks)**

b) Explain, the main phases of network design as per the PDIOO approach. **(08 marks)**

c) Explain why it is important to understand ***Kopala University's*** business style when designing their network. **(06 marks)**

d) Describe what tradeoffs may be necessary in order to improve ***Kopala University***'s network efficiency. **(04 marks)**

e) Evaluate the difference between a router and a switch in network design. **(04 marks)**

f) During your design process, you design the following as part of your network topology;

Topology Diagram

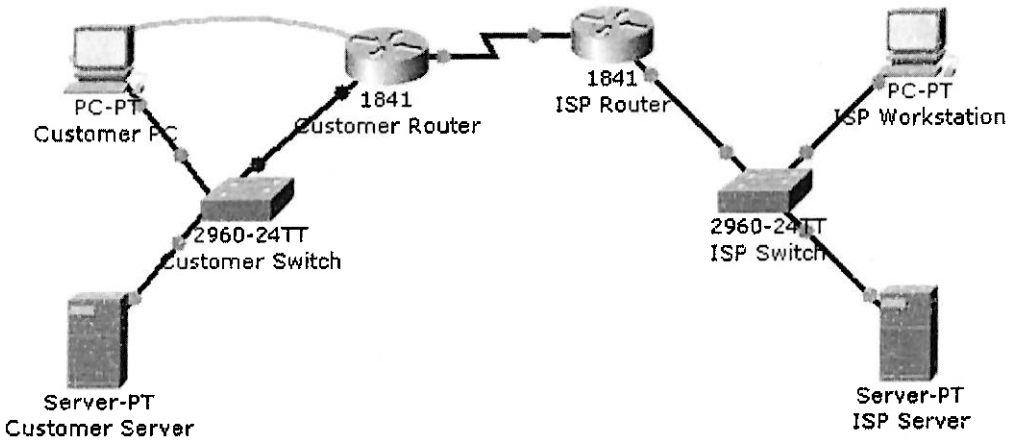


Figure 1.

Configure the **Customer Router** with the following settings;

a) Hostnme: **"Kopala1"**

b) Banner: **"No unauthorized login attempts on Kopala University Network"**

192.168.2.1

c) Assign: IP address **192.168.2** to interface **serial 0/0/0** and IP address **172.16.1.3** to **fast Ethernet 0/0**

d) Assign: Clock Rate of **64000kbps** to **Serial 0/0/0** **(12 marks)**

(Total marks for Section A: 40 marks)

SECTION B

CHOOSE ANY THREE (03) QUESTIONS FROM THIS SECTION

QUESTION TWO

CASE STORY: *Peter, David and Mark*

Three researchers, ***Peter***, ***David*** and ***Mark*** are given the task to create a new web search engine similar to Google. Just like the Google web search engine, they are required to design a search engine that handles large volumes of data and spans several internetworks. You are a student studying Network Design and Security at ZCAS University and you are tasked to help these three researchers by answering the following questions so as to aid them create a web search engine that meets the network requirements:

a) Briefly describe how network redundancy ensures business continuity for 24/7 broadcasting operations. **(04 marks)**

b) Explain the difference between effectivity and efficiency? **(04 marks)**

c) Discuss **TWO (02)** strategies a network designer can use to ensure scalability and two strategies to enhance network performance. **(04 marks)**

d) Describe the **THREE (03)** main types of internetwork that are used to support distributed systems. In your answer refer to standard networks and the wireless variants of them. **(06 marks)**

e) What role does data encryption play in protecting client information during transmission and storage? **(02 marks)**

(Total: 20 marks)

QUESTION THREE

CASE STORY: *Punzila University*

Punzila University is an award-winning tier-one tertiary institution that offers short-term and long-term programmes in business management, ICT and Consultancy. Strategically located in Mkushi, Zambia, the university accommodates students from diverse regions who benefit from its state-of-the art campus network. The university has a secured campus network, which

includes student dormitories, faculty offices, and computer labs.

You are hired as a Network Engineer for ***Punzila University*** and you are tasked with implementing security policies to protect sensitive student data.

a) Explain the importance of a Demilitarized Zone (DMZ) in securing externally-facing services such as the university's web servers. **(4 marks)**

b) Compare and contrast Layer **2** and Layer **3** network devices in the context of segmentation and traffic control. **(04 marks)**

c) Explain the importance of network security protocols in network design. **(02 marks)**

d) Identify and explain **TWO (02)** security measures you would implement to protect this network. **(04 marks)**

e) During your network management, you have detected a network security breach. Describe the steps you would take to respond to this incident and mitigate its impact. **(06 marks)**

(Total: 20 Marks)

QUESTION FOUR

CASE STORY: ***Magoye General Hospital***

The ***Magoye General Hospital*** is redesigning its network to improve patient data security and compliance with healthcare regulations. Also, they have deployed IoT devices for patient monitoring and smart ward management. The hospital administration is concerned about securing this highly connected environment. There are several advices they have received concerning the security issue. As their Network Design and Security Expert, you are tasked to provide guidance on the following concerns;

a) Explain the security challenges associated with integrating IoT devices into a hospital network? **(04 marks)**

b) Explain the role of MAC address filtering in enhancing network access control. **(04 marks)**

c) Differentiate between preventive and detective security controls with relevant examples. **(04 marks)**

d) Explain the role NAT play in network design and security? **(04 marks)**

e) Evaluate the concept of network segmentation and why it is important in network design. **(04 marks)**

(Total: 20 Marks)

QUESTION FIVE

CASE STORY: ***The National Revenue Authority***

The National Revenue Authority plans to interconnect its regional offices and enable secure access to its tax processing system. To achieve this, they have sought your expert advice on the network design and security aspects.

a) Explain how the use of Dynamic Host Configuration Protocol (DHCP) simplifies IP address management in large networks. **(04 marks)**

b) What is the importance of including network logs in the overall security monitoring strategy? **(04 marks)**

c) Differentiate between wired and wireless networks in terms of security vulnerabilities and how to mitigate them. **(04 marks)**

d) Explain the purpose of including a firewall in network design? **(04 marks)**

e) Compare IDS with IPS (IPS) in terms of function and deployment. **(04 marks)**

(Total: 20 marks)

END OF EXAMINATIONS